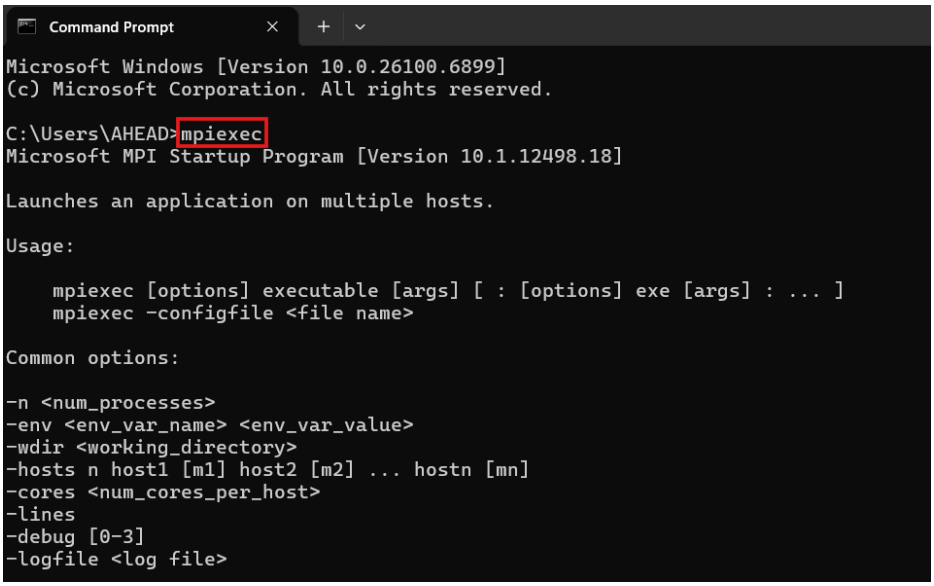
Setup Instructions for running MPI Program

1. Install Visual Studio
    1. Download from: https://visualstudio.microsoft.com/downloads/
    2. Choose Community (free).
    3. Select workload: Desktop development with C++
2. Download and Install MS-MPI (**Microsoft MPI)**
    1. Download from:
        1. Official site: https://learn.microsoft.com/en-us/message-passing-interface/microsoft-mpi
        2. Direct: https://www.microsoft.com/en-us/download/details.aspx?id=105289
    2. Install both files (run as Administrator):
        1. Runtime: MSMpiSetup.exe (or msmpisetup.exe)
        2. SDK: msmpisdk.msi (for headers & libs)
        3. Default paths:
            1. Bin: C:\Program Files\Microsoft MPI\Bin
            2. Include: C:\Program Files (x86)\Microsoft SDKs\MPI\Include
            3. Lib (x64): C:\Program Files (x86)\Microsoft SDKs\MPI\Lib\x64
        4. Make sure that installation of MS-MPI is completed via cmd:

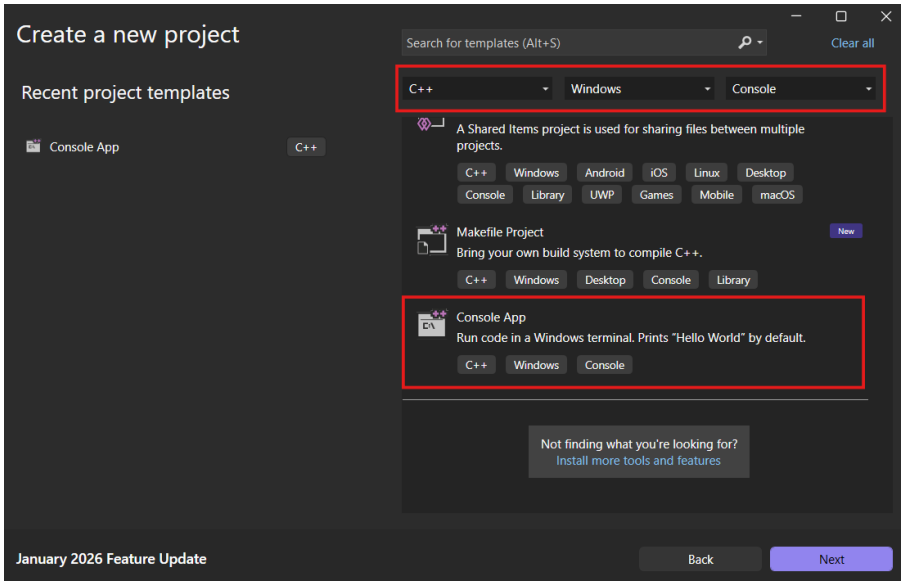


If installation is completed, you will get an output like in the image above. If not, set the path of System variables' path from Environment variables using MS-MPI's Bin folder path.

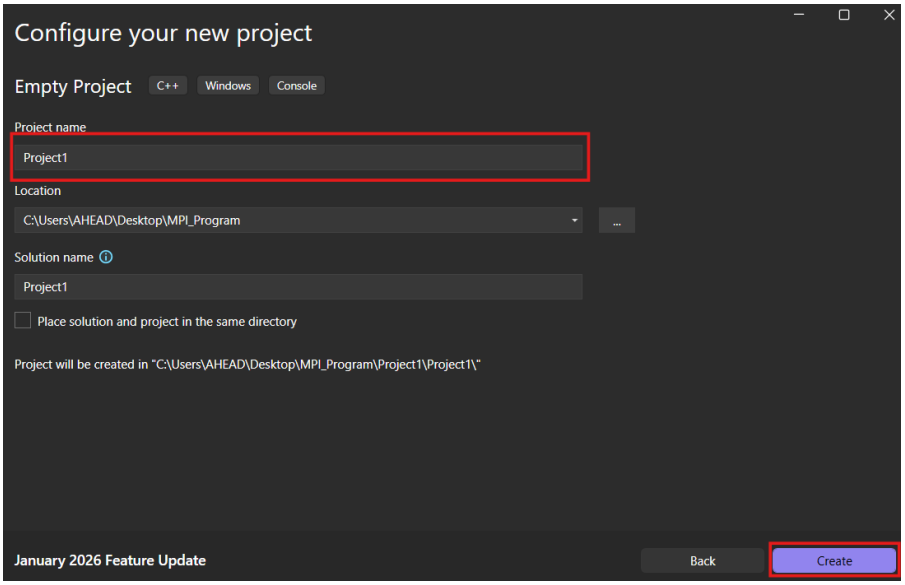
Bin: C:\Program Files\Microsoft MPI\Bin

3. Create and Configure Project Properties (Include Path)

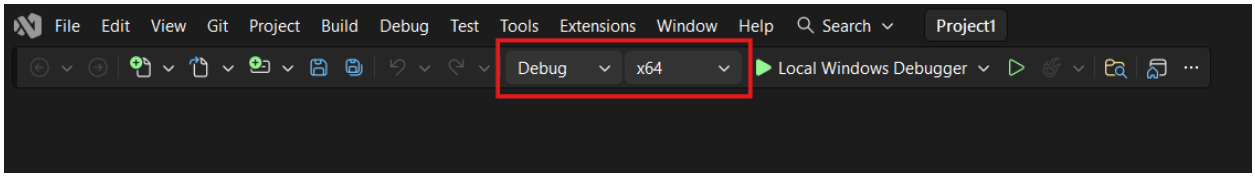
   1. Select Language option as **"C++"**, Platform and Project type as **"Console"**, under shown option click **"Console App"** then click **"Next"** button.

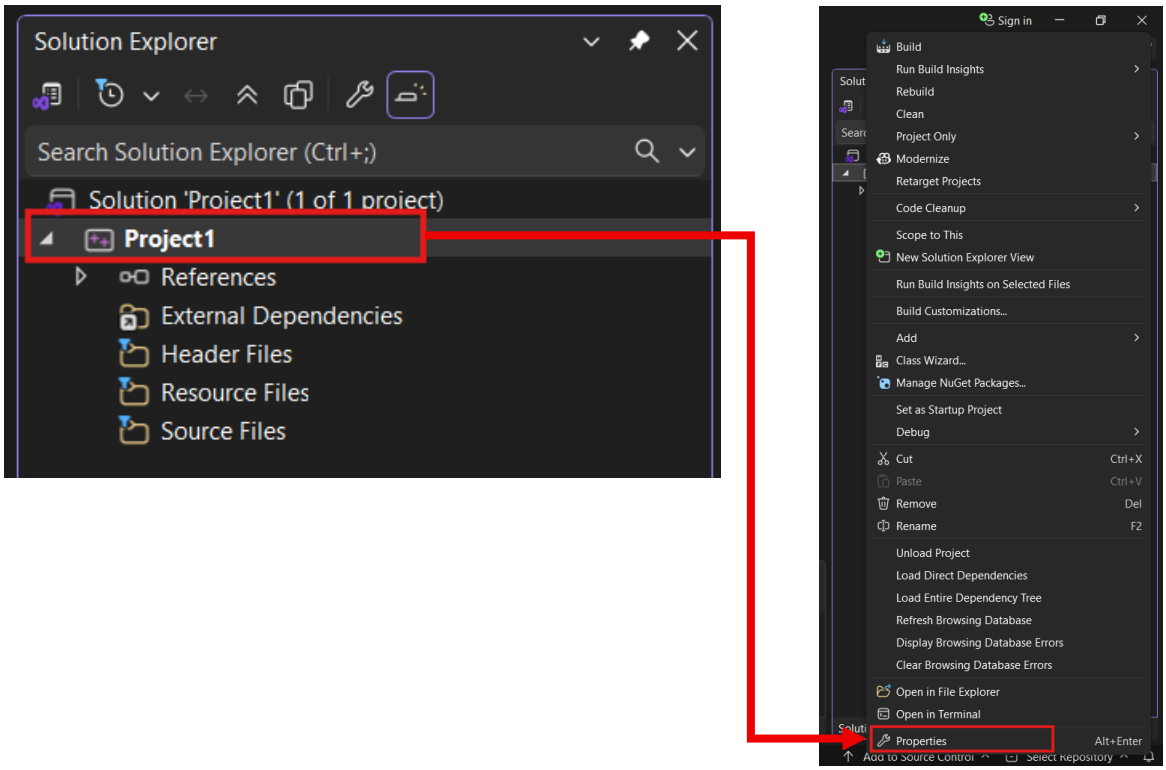


   After that, name you application as you want and click the **"Create"** button to create application in specified location.



2. First set the Program Platform as: **“x64”** Next, Right click on project then move to the option **“Properties”**

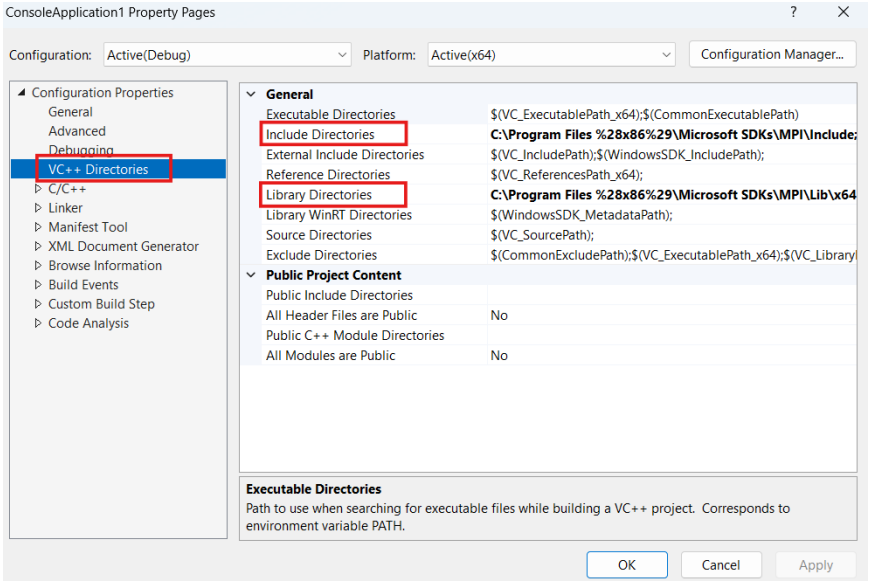




3. Form Pop-up screen of **“Properties”**, Click the option **“VC++ Directories”** and set the paths for selected options under General Categories **“Include Directories”** and **“Library Directories”** which are boxed in red color in Images below. We have installed two files for MS-MPI, which are **“Microsoft MPI”** and **“Microsoft SDKs”,** where you would have found the paths for subfolders **“include”** and **“lib”**. Next, we are going to set those paths that selected options of **“VC++ Directories”**.

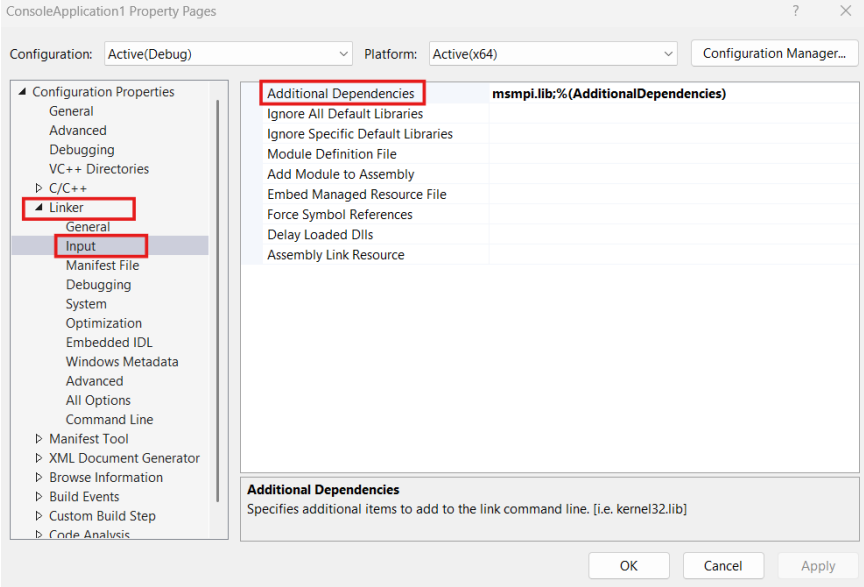
    1. Include Directories: **“C:\Program Files (x86)\Microsoft SDKs\MPI\Include”**
    2. Library Directories: **“C:\Program Files (x86)\Microsoft SDKs\MPI\Lib\x64”**

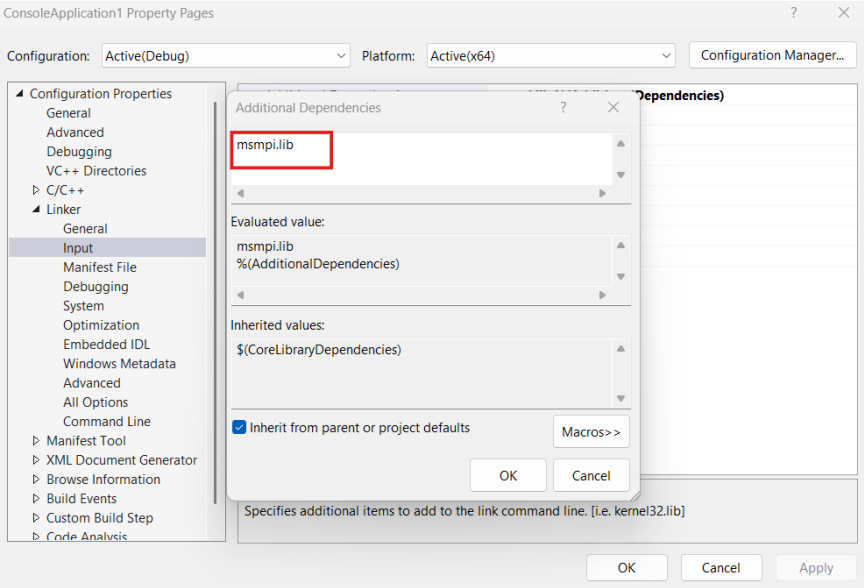
    Note: The given paths are sample ones for your understanding.



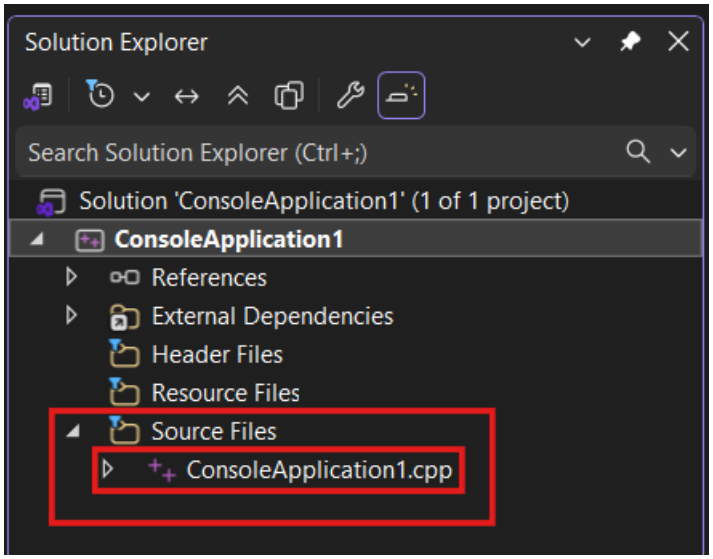
4. Select **"Linker"** option and under that choose the categories **"Input"** from that click option **"Additional Dependencies"**, where we need to type the following text: **msmpi.lib**

   After complete whole steps for project configuration, Click the button Apply to updates project.





4. Create a Simple MPI Project in Visual Studio, Generally Visual Studio creates sample program file inside boxed option **"Source Files"**, Create program file with extension of **.cpp**.



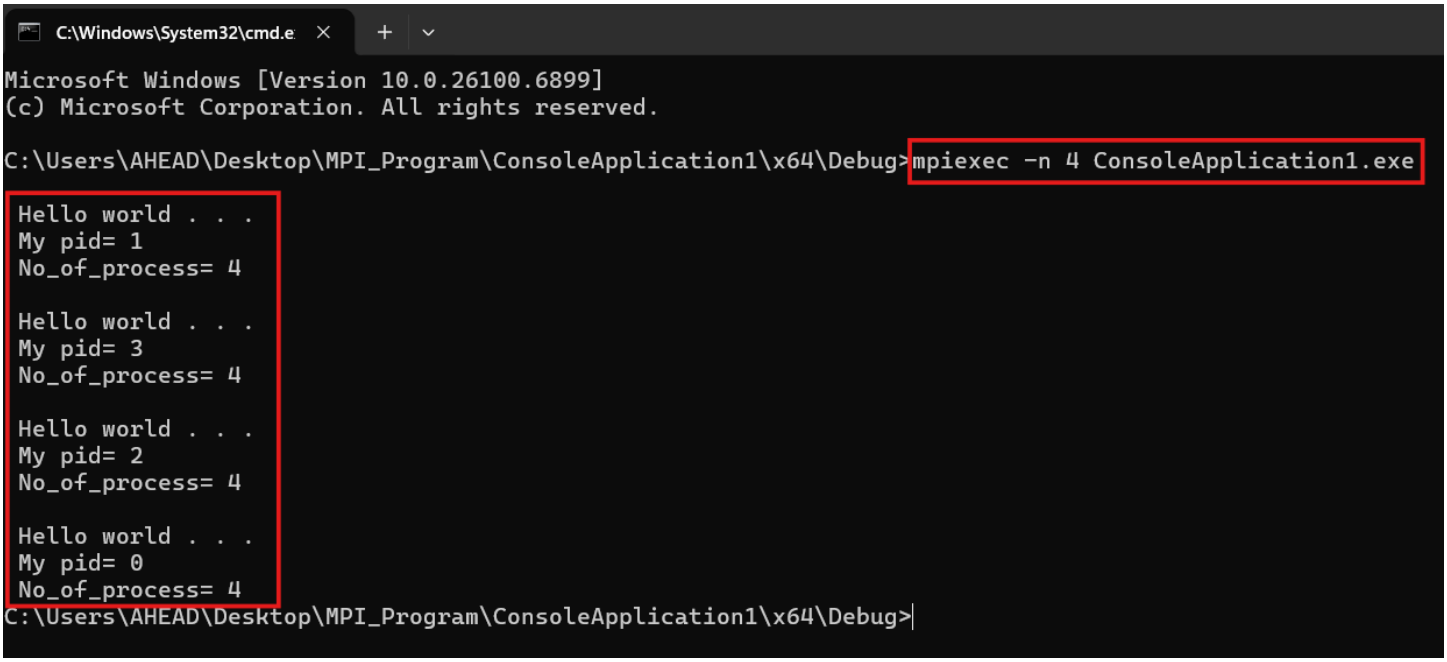
1. Type the Given program inside the created file, after typed whole program, first you need to Build your application to create executable file for this application: **.exe**. then only we can run this application.

   To Build Application: **Ctrl+Shift+B**

```
#include <mpi.h>
#include <stdio.h>
int main()
{
        int np;
        int pid;
        MPI_Init(NULL,NULL) ;
        MPI_Comm_size(MPI_COMM_WORLD,&np);
        MPI_Comm_rank(MPI_COMM_WORLD,&pid);
        printf("\n Hello world . . . \n My pid= %d \n No_of_process= %d",pid,np);
        MPI_Finalize();
return 0;
}
```

5. After build the application, make sure that executable has been created in the location below. Then open Command prompt(cmd) from this path to run your application.

   C:\Users \Desktop\MPI_Program\ConsoleApplication1\x64\Debug

   Note: This is sample path for your references.

   To run your application: **mpiexec -n No_of_Processors ConsoleApplication1.exe**